For Examiner's Use

2 Fig. 2.1 shows the variation with distance from the centre of the Earth of the gravitational field strength *g*.

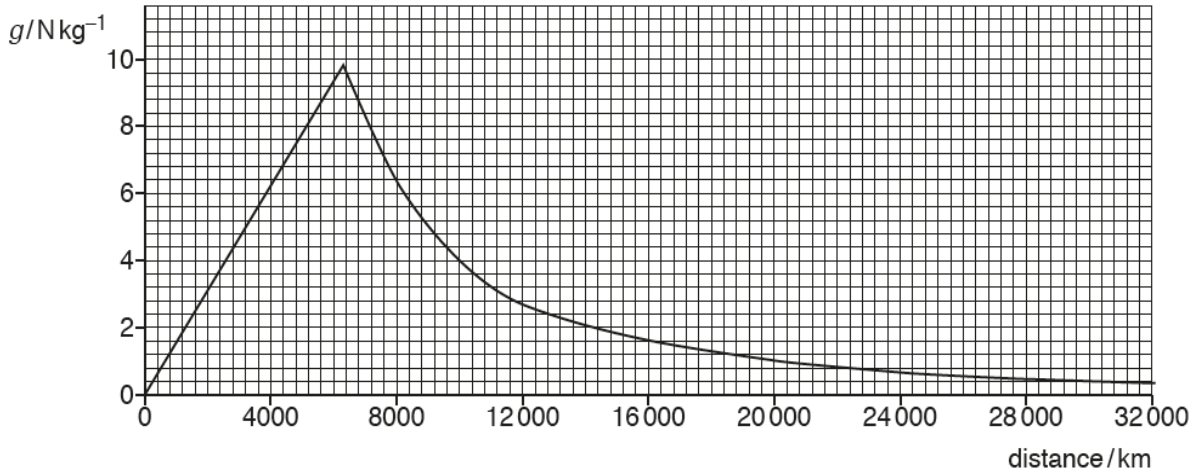

Fig. 2.1

(a) Use Fig. 2.1 to determine the gravitational force on a man-made satellite of mass 20 000 kg at a distance of 8200 km from the centre of the Earth.

gravitational force = ... N [2]

(b) Calculate the speed of the satellite in **(a)** for it to be circling the Earth at constant speed.

speed = ... m s^{-1} [2]

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(c) **(i)** State what is meant by *gravitational potential*.

……

…… [1]

(ii) Use Fig. 2.1 to estimate the gravitational potential at a distance of 10 000 km from the centre of the Earth.

gravitational potential = ... J kg^{-1} [2]